

PHOTODIODE-BASED SUN SENSOR FOR CUBESAT ATTITUDE DETERMINATION

Devin Zhang

ABSTRACT

This paper presents the design, implementation, and testing of a photodiode-based sun sensor system for CubeSat attitude determination. The system utilizes an array of six photodiodes mounted on each face of a cube to detect light intensity, enabling the determination of a light vector corresponding to the direction of an external light source relative to the CubeSat. The data from the photodiodes is processed through an Analog-to-Digital Converter (ADC) connected to a Raspberry Pi 4, allowing for real-time vector calculation and visualization. This work is part of a larger Attitude Determination and Control System (ADCS) for a CubeSat, which combines the sun sensor with a magnetometer to implement a two-vector attitude determination algorithm. Testing demonstrated the system's ability to accurately track a light source, though challenges including ambient light interference and signal jitter were encountered and addressed. The developed prototype successfully demonstrates the viability of low-cost photodiode-based sun sensors for small satellite applications, laying groundwork for future refinements toward a flight-ready system.

NOMENCLATURE

ADCS	Attitude Determination and Control System
ADC	Analog-to-Digital Converter
ECI	Earth centered inertial reference frame
ECEF	Earth centered earth fixed reference frame
PWM	Pulse Width Modulation
SPI	Serial Peripheral Interface
TRIAD	TRIaxial Attitude Determination
WMM	World Magnetic Model
LSB	Least Significant Bit

1. INTRODUCTION

CubeSats have revolutionized space exploration by providing a standardized, low-cost platform for small satellite missions. A critical component of any satellite is its Attitude Determination and Control System (ADCS), which enables the satellite to determine its orientation in space and adjust it as needed for mission objectives. For CubeSats, which face strict constraints on size, weight, and power consumption, developing efficient and reliable ADCS components is particularly challenging. This project focuses on the development of a photodiode-based sun sensor system, a crucial element of the ADCS for determining the satellite's orientation relative to the sun. Sun sensors are widely used in spacecraft due to their simplicity, reliability, and the sun's prominence as a reference point in space [1]. By accurately measuring the direction of sunlight, these sensors provide essential data for satellite orientation calculations. The Attitude Determination and Control System is one of the most important components of a satellite because

every aspect of a satellite's mission, from its payload to its power generation to its communication systems, requires the ability to determine the position and orientation of the satellite [2].

The goal for this semester was to design, implement, and test a functional photodiode-based sun sensor system that could determine the direction of a light source and integrate this data with a magnetometer for complete attitude determination using a two-vector method. The approach involved mounting photodiodes on each face of a cubic structure, developing algorithms to process the intensity readings, and implementing visualization tools to demonstrate the system's effectiveness. The system must be accurate to within $\pm 5^\circ$ and be cost-effective, which led to the decision to use simple photodiode-based sun sensors rather than more complex alternatives [2].

In the subsequent sections, I will detail the technical background of sun sensors and attitude determination, describe the design and implementation process, present testing procedures and results, and discuss future improvements for the system.

2. BACKGROUND

2.1 Sun Sensors in CubeSat Applications

Sun sensors are among the most common sensors used in spacecraft attitude determination systems due to their reliability and the sun's role as a consistent reference point [2]. In CubeSats, where resources are limited, simple yet effective sun sensor designs are particularly valuable.

Traditional sun sensors can be categorized into analog and digital types. Analog sun sensors, like the photodiode-based system developed in this project, measure the intensity of sunlight hitting sensors positioned at different angles. Digital sun sensors use more complex optical systems with position-sensitive detectors or imaging arrays [3]. For small CubeSats, analog sensors offer advantages in terms of simplicity, power consumption, and cost-effectiveness. The specific photodiode model chosen for this project (SFH 2430-Z) has been successfully used in previous CubeSat missions for sun sensing applications [2], providing confidence in our design approach.

2.2 Two-Vector Attitude Determination

Attitude determination requires at least two reference vectors to fully define a satellite's orientation in three-dimensional space. Common vector pairs used in CubeSats include sun vector/magnetic field vector, sun vector/earth vector, and star tracker/horizon sensor combinations [4].

The TRIAD algorithm, which was implemented in this project, is a deterministic method that uses two vector measurements to determine the attitude quaternion. This algorithm transforms the body-frame measurements into an inertial reference frame, allowing for the calculation of the satellite's orientation [5]. While not the most computationally efficient algorithm

(compared to more advanced methods like QUEST or Davenport's q-method), TRIAD provides a straightforward implementation suitable for prototype development.

It's important to note that a single sensor (such as a sun sensor or magnetometer) cannot provide enough information to fully determine the satellite's attitude. This is because each sensor produces a single vector, and knowing the orientation of the satellite with respect to that vector still allows the satellite to be rotated about that vector by any angle. Only by combining two vectors, such as the sun and magnetic vectors, can the satellite's attitude be fully determined [2].

2.3 Reference Frames and Coordinate Systems

For rigorous work with attitude determination, it is necessary to have a fixed set of reference frames. Two earth frames are commonly defined: Earth Centered Inertial (ECI) and Earth Centered Earth Fixed (ECEF). The ECI reference frame is fixed to the position of the earth but does not rotate with it. The z-axis points toward the north pole, the x-axis points toward the vernal equinox, and the y-axis completes the right-hand orientation of the coordinate system [2].

The ECEF frame is also fixed to the earth but rotates with it, with its x-axis pointing toward the prime meridian. Both frames are important for satellite operations and attitude determination. The ECI frame is necessary because satellite orbits are determined in this frame as they are independent of the earth's rotation. The ECEF frame is necessary because the orientation of the earth's magnetic field depends on the rotation of the earth [2].

In addition to these earth-centered reference frames, a body-fixed reference frame attached to the satellite is used. The objective of the attitude determination system is to obtain the rotation matrix between these reference frames.

2.4 Photodiode Fundamentals

Photodiodes are semiconductor devices that convert light into electrical current. When exposed to light, they generate a current proportional to the light intensity hitting their surface. This property makes them ideal for sun sensing applications where the goal is to determine the direction of incident light. In CubeSat applications, photodiodes must be carefully selected considering factors such as spectral response, operating temperature range, and radiation tolerance [6]. For this project, standard photodiodes were used in conjunction with resistors to create a voltage divider circuit, allowing voltage measurements proportional to light intensity.

The intensity of light detected by a photodiode varies with the cosine of the angle between the incoming light and the photodiode's normal direction. Because of this property, when photodiodes are mounted in mutually perpendicular directions, the light vector can be determined in the photodiodes' frame of reference by assigning the current output of each photodiode to the component of the light vector in that photodiode's normal direction [2].

As shown in Figure 1, two photodiodes can be combined to determine a sun vector component in a single plane. By using multiple such pairs or a full array of photodiodes covering

different orientations, a complete three-dimensional sun vector can be determined.

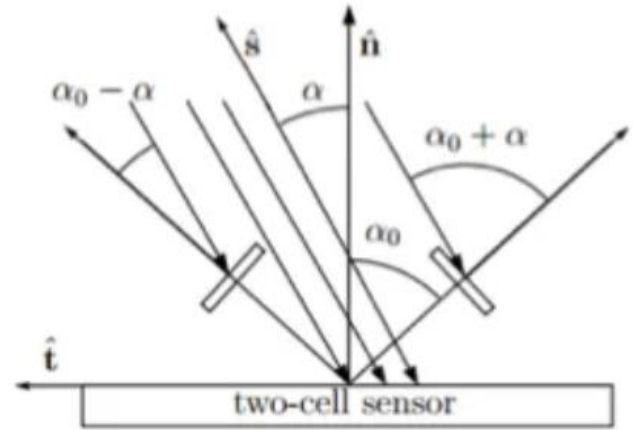


Figure 1: Two Photodiodes combined to get Sun Vector Component in a Single Plane

For a reverse-biased photodiode configuration, which provides fast response times, high sensitivity, and good signal-to-noise ratio, the voltage measured across a resistor is directly proportional to the photocurrent generated when light hits the photodiode ($V = i \times R$). According to the datasheet for our photodiodes, with a 5V reverse bias and 1000 lux illumination, the expected photocurrent ranges from 5.6 μA to 6.3 μA .

3. DESIGN

3.1 System Architecture

The sun sensor system consists of hardware and software components designed to work together for accurate light vector determination. The system architecture, shown in Figure 2 comprises six photodiodes mounted on each face of a cubic structure, an Analog-to-Digital Converter (MCP3008), a Raspberry Pi 4 for processing, and visualization script for real-time feedback.

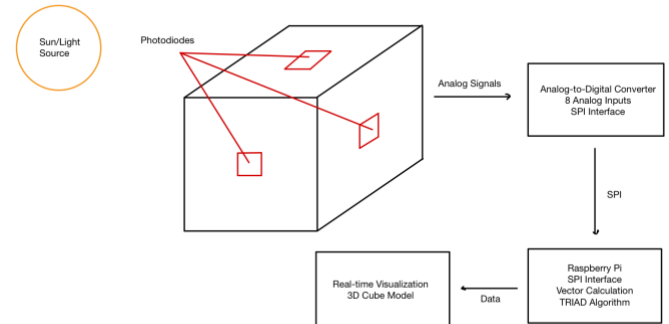


Figure 2: System Architecture Schematic

The design principles guiding this work included:

- **Simplicity:** Using readily available components and straightforward construction techniques
- **Accuracy:** Optimizing sensor placement and signal processing for reliable vector determination

- Modularity: Ensuring the sun sensor could be integrated with other ADCS components
- Educational value: Creating a system that demonstrates fundamental principles of attitude determination

3.2 Hardware Implementation

Photodiode Circuit Design

Each photodiode was connected in a voltage divider configuration with a $10\text{k}\Omega$ resistor, as shown in Figure 3. This arrangement produces a voltage proportional to the light intensity hitting the photodiode, which can be measured by the ADC. The specific circuit was chosen for its simplicity and adequate sensitivity to light changes.

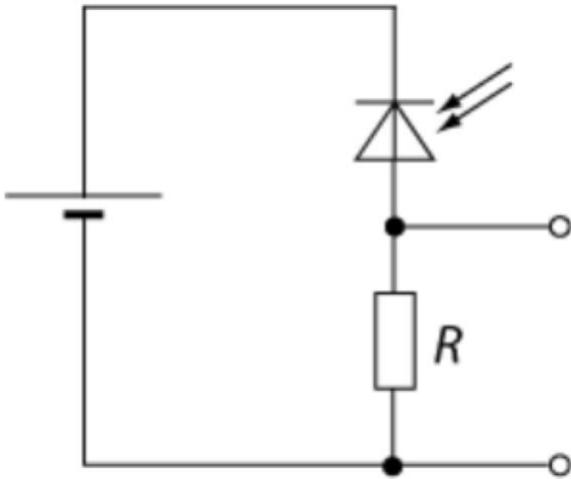


Figure 3: Photodiode Circuit

Initial testing revealed that the $10\text{k}\Omega$ resistor provided a good balance between sensitivity and safe voltage levels for the ADC. Higher resistance values would increase sensitivity but might risk exceeding the ADC's maximum input voltage ($3.3\text{V} + 0.6\text{V}$) when exposed to intense light.

Through experimentation, I characterized the photodiode behavior under various lighting conditions using a multimeter connected in series between the photodiode and resistor:

- Blocked photodiode: $0\text{ }\mu\text{A} \rightarrow 0.02\text{V}$ across the $10\text{k}\Omega$ resistor
- Ambient lighting: $2\text{ }\mu\text{A} \rightarrow 0.02\text{V}$
- Flashlight at 11": $55\text{ }\mu\text{A} \rightarrow 0.55\text{V}$
- Flashlight at 8.5": $90\text{ }\mu\text{A} \rightarrow 0.9\text{V}$
- Flashlight right up front: $289\text{ }\mu\text{A} \rightarrow 2.89\text{V}$

These measurements confirmed that the circuit design provided adequate resolution for detecting varying light intensities while staying within the safe operating range of the ADC.

ADC Integration

The MCP3008 10-bit ADC was selected for converting the analog voltage signals from the photodiodes to digital values readable by the Raspberry Pi. This ADC communicates with the

Raspberry Pi via the SPI protocol, allowing for multiple analog inputs to be read using a single chip.

The physical integration involved mounting the ADC on a breadboard and connecting it to the Raspberry Pi's GPIO pins for power (3.3V) and SPI communication. Each photodiode circuit was connected to one of the eight analog input channels of the ADC.

The ADC specifications were carefully considered for optimal performance:

- Operating voltage (VDD): $2.7\text{--}5.5\text{V}$
- Input voltage constraint: must be $< \text{VDD} + 0.6\text{V}$ to avoid damage
- Reference voltage (Vref): determines the upper limit of input voltage measurement
- Resolution: 10-bit (0-1023), giving a least significant bit value of 3.22mV with $\text{Vref} = 3.3\text{V}$

Although lowering VDD to 2.7V could improve resolution by approximately 18% (reducing LSB to 2.64mV), we determined that the standard 3.3V configuration provided sufficient accuracy for our application without requiring additional circuit modifications.

3.3 Structural Design

The photodiodes were mounted on each face of a cubic structure to achieve full spherical coverage for light detection. Initially, a temporary structure was created using zip ties and tape for concept validation. Later, the photodiodes were integrated into a more robust laser-cut wood cube designed to be mounted on a gimbal for testing, as shown in Figure 4.

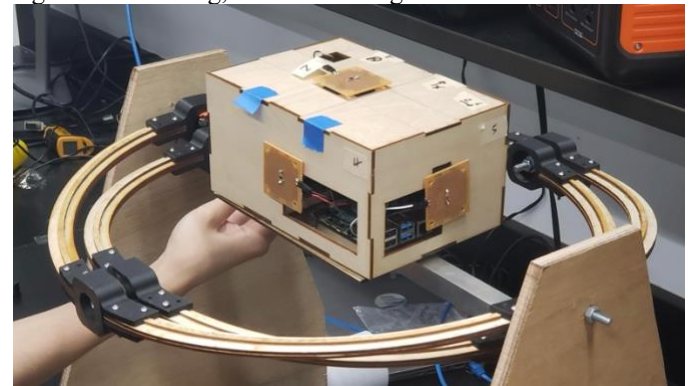


Figure 4: Laser-cut Wooden Cube and Gimbal

This design maximized the number of photodiodes that could be exposed to a light source at any given orientation, ensuring adequate data for vector calculation. However, one challenge encountered was that the photodiode on the face attached to the gimbal had limited exposure to light, affecting its readings.

For the initial testing phase, I started with three photodiodes mounted orthogonally to get readings on each axis. As the project progressed, this was expanded to six photodiodes (one on each face of the cube) to achieve complete coverage of all possible light directions. This configuration was inspired by the design used in the RAX-2 CubeSat mission, though their implementation used three photodiodes on each side for 99.5% coverage of the attitude sphere [3].

3.4 Software Implementation

Vector Calculation Algorithm

The core algorithm for light vector determination processes the intensity readings from all six photodiodes to calculate a normalized vector pointing toward the light source. The algorithm consists of three main steps:

1. **Data acquisition:** Reading voltage values from each photodiode through the ADC
2. **Vector calculation:** Converting intensity readings into a directional vector using the following approach:
 - For each axis (x, y, z), calculate the difference between opposing faces
 - Normalize the resulting vector to produce a unit vector pointing toward the light source
3. **Vector filtering:** Implementing a low-pass filter to reduce noise and jitter in the calculated vector

The mathematical implementation for the vector calculation is as follows:

```
# Converting ADC readings to a directional vector
x_component = +x_face_reading - -x_face_reading
y_component = +y_face_reading - -y_face_reading
z_component = +z_face_reading - -z_face_reading

# Creating the raw vector
raw_vector = [x_component, y_component, z_component]

# Normalizing the vector
magnitude = sqrt(sum([component**2 for component in
raw_vector]))
if magnitude > 0:
    normalized_vector = [component/magnitude for component
in raw_vector]
else:
    normalized_vector = [0, 0, 0]
```

This approach assumes that the photodiodes on opposite faces will have different readings when illuminated by a directional light source, with the difference proportional to the light's direction. The algorithm is based on the principle that the intensity detected by a photodiode varies with the cosine of the angle between the incoming light and the photodiode's normal direction [2].

For later iterations of the algorithm, I improved it to select the three photodiodes with the highest intensity readings to calculate the sun vector, as these would be the faces most directly exposed to the light source:

```
# Find the three diodes with highest intensities
sorted_indices = sorted(range(6), key=lambda i: readings[i],
reverse=True)
top_three = sorted_indices[:3]
# Use these three to calculate the vector
```

This approach proved more robust in certain lighting conditions, especially when ambient light was a factor.

3.5 Visualization System

A real-time visualization system was developed using Python and Matplotlib to provide immediate feedback on the calculated light vector. The visualization displays:

1. A 3D representation of the cube with its current orientation
 2. The calculated light vector as an arrow pointing from the center of the cube
 3. Individual intensity readings visualized as shorter vectors on each face, with magnitudes proportional to the readings
- This visualization proved invaluable for debugging and demonstrating the system's functionality, allowing for direct observation of the relationship between physical movement of the light source and the calculated vector.

For the integrated system with both photodiode and magnetometer data, the visualization was expanded to show the calculated attitude using the TRIAD algorithm. However, performance issues were encountered with the visualization running on the Raspberry Pi, leading to low frame rates that made real-time assessment difficult. This was partially addressed by upgrading to a faster SD card for the Pi, but further optimization of the visualization code would be beneficial for future iterations.

3.6 Integration with Magnetometer

To achieve complete attitude determination, the sun sensor was integrated with a magnetometer (AK09940) that had been previously configured. The TRIAD algorithm was implemented to combine the light vector (simulating the sun vector) and the magnetic field vector to calculate the full attitude quaternion.

The software integration required:

- Synchronizing data collection from both sensors
- Transforming the vectors into compatible coordinate systems
- Implementing the TRIAD algorithm for quaternion calculation
- Updating the visualization to display the calculated attitude

The integration of these two sensors is based on the principle that both the sun vector and magnetic field vector can be determined in two different reference frames. In the body-fixed frame, these vectors are measured directly by the sensors. In the ECI/ECEF frames, these vectors can be calculated using astronomical models and geomagnetic models.

For the magnetic field vector in the ECEF frame, we used the World Magnetic Model (WMM) maintained by NOAA. This model provides the direction and magnitude of Earth's magnetic field at any latitude, longitude, and altitude, with directional accuracy within 1° up to 4,000 km above sea level [2]. This is more than adequate for our low Earth orbit application.

For the sun vector in the ECI frame, we used the Python library Poliastro, which provides functionality for determining the position of celestial bodies, including the sun, in the Geocentric Celestial Reference System (equivalent to ECI) [2]. This allowed us to calculate the theoretical sun vector for any given time and location.

The TRIAD algorithm then used these four vectors (two measured in the body frame and two calculated in the reference

frames) to determine the rotation matrix that describes the attitude of the satellite.

4. TESTING AND RESULTS

4.1 Test Setup

Multiple testing procedures were developed to validate the sun sensor's performance:

- **Basic functionality testing:** Using a flashlight to illuminate different faces of the cube and observing the calculated vector
- **Noise analysis:** Measuring the stability of readings with a stationary light source
- **Accuracy assessment:** Comparing calculated vectors with known light source positions
- **Integration testing:** Verifying the combined operation of the sun sensor and magnetometer

The test setup consisted of the photodiode cube mounted on a gimbal for controlled movement, a flashlight serving as the light source, and a computer running the visualization software. For some tests, a more powerful light source (Ossila Solar Simulator) was used to better simulate sunlight intensity. The solar simulator proved particularly effective as it provided a much brighter and more consistent light source than standard flashlights, helping to overcome the ambient light interference issues.

For the TRIAD algorithm testing, I set up a stationary light source (using my phone's camera flash) to act as a fixed inertial frame sun vector. The inputs to the algorithm were:

- `sun_inertial`: stationary solar light direction
- `sun_body`: photodiode readings
- `mag_inertial`: IGRF model at Cooper Union's coordinates
- `mag_body`: magnetometer readings (or simulated constant vector for initial testing).

4.2 Results and Analysis

Basic Functionality

Initial testing confirmed that the system could detect a light source and calculate a reasonable vector pointing toward it. When illuminating different faces of the cube, the visualization showed the vector adjusting appropriately, demonstrating the basic functionality of the sensor system.

Signal Quality and Noise

Analysis of signal quality revealed several challenges:

- **Ambient light interference:** Initial tests showed significant influence from room lighting, affecting the accuracy of the calculated vector. This was temporarily mitigated by covering the setup with a towel, but highlighted the need for better optical filtering.
- **Signal jitter:** Unexpected fluctuations in readings were observed even with a stationary light source, as shown in Figure 5. Oscilloscope analysis revealed periodic dips in the voltage readings.

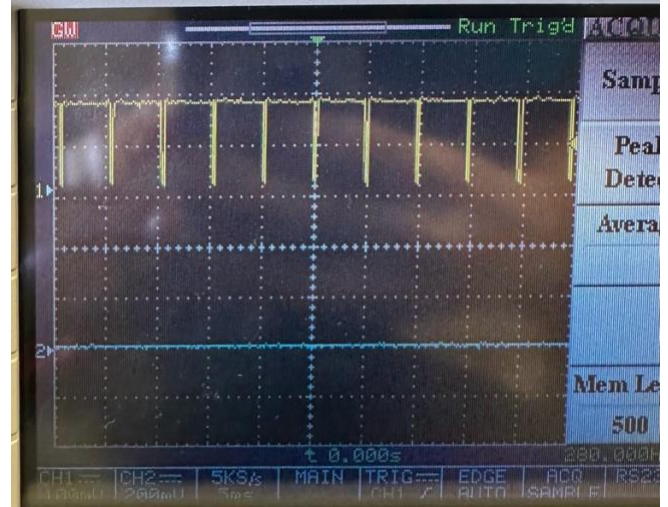


Figure 5: Oscilloscope Analysis

Further investigation determined that the jitter was caused by:

- PWM (Pulse Width Modulation) in the flashlight, causing imperceptible flickering that was detected by the photodiodes
- Possible digital switching noise from the Raspberry Pi affecting the analog signals

To address these issues, 0.1 μ F capacitors were added in parallel with each resistor to act as low-pass filters, reducing the impact of high-frequency fluctuations. This modification improved stability but did not completely eliminate the jitter.

One simple test to confirm the PWM effect was pointing a smartphone camera at the flashlight, which revealed vertical lines in the camera view – a clear indication of PWM flickering that is imperceptible to the human eye but detectable by both the camera and the photodiodes [3].

Interestingly, when testing with a different, smaller flashlight, the jittering effect was not observed, suggesting that the choice of light source can significantly impact the quality of the measurements. However, the smaller flashlight was not bright enough to overcome ambient lighting effects without being positioned very close to the photodiodes, which would create non-parallel light rays and complicate the vector calculation.

Another issue encountered was an occasional voltage offset in all photodiodes, where all readings would suddenly show approximately 1V, but still respond correctly to changes in light direction. This issue seemed to resolve itself without clear cause, suggesting possible grounding or wiring issues that would need to be addressed in a more robust design.

4.3 Accuracy Assessment

To assess accuracy, the light source was positioned at known angles relative to the cube, and the calculated vector was compared to the expected vector. While precise quantitative measurements were difficult with the available equipment, qualitative assessment showed that the system could detect changes in light direction with reasonable accuracy.

The primary accuracy limitations identified were:

- Inconsistent sensitivity between different photodiodes

- Ambient light effects
- Reflections within the test environment affecting readings

These findings align with research by Adin Sacho-Tanzer, who conducted a characterization test for photodiodes in a similar CubeSat project. Their testing showed that when the photodiode was pointed away from the primary light source, it still received light reflected from walls and other surfaces, leading to unexpected non-zero outputs [7]. This effect is analogous to the Earth albedo (reflected sunlight) that will affect readings in the space environment.

For future iterations, on-orbit, attitude-independent photodiode calibration methods could be implemented. This approach uses a minimization problem to estimate the peak voltage output of each sensor, which can be formulated as a linear least-squares problem [3]. Such calibration has been successfully applied to real CubeSat missions like RAX-1.

4.4 Integration with Magnetometer

Integration testing with the magnetometer and TRIAD algorithm showed promising results. The combined system could determine the cube's orientation when both sensors provided reliable data. However, several challenges were encountered:

1. Difficulty in simulating a controlled magnetic field for thorough testing
2. Alignment issues between the coordinate systems of the two sensors
3. The need for careful calibration of both sensors to achieve accurate attitude determination

Despite these challenges, the integrated system demonstrated the viability of the two-vector approach for attitude determination in the prototype CubeSat.

5. Conclusion and Future Work

This project successfully designed, implemented, and tested a photodiode-based sun sensor system for CubeSat attitude determination. The system demonstrated the ability to detect a light source's direction using simple, low-cost components, and was successfully integrated with a magnetometer to implement a complete attitude determination algorithm.

The primary contributions of this work include:

1. A functional photodiode array design for omnidirectional light sensing
2. Real-time vector calculation and visualization software
3. Integration with magnetometer data for TRIAD algorithm implementation
4. Identification and partial mitigation of key challenges in photodiode-based sensing

While the prototype system achieved its basic goals, several areas for improvement have been identified for future work:

5.1 Sensor Enhancements

1. **Optical filtering:** Implementing spectral bandpass filters to isolate sunlight and reduce ambient light interference

2. **Improved physical design:** Developing a more compact and robust housing with better photodiode alignment
3. **Circuit optimization:** Redesigning the photodiode circuits with better noise immunity, possibly using operational amplifiers for signal conditioning
4. **Custom PCB integration:** Moving from breadboard to a printed circuit board design for improved reliability and reduced noise

5.2 Algorithm Improvements

1. **Advanced filtering:** Implementing more sophisticated digital filters to reduce noise and improve vector stability
2. **Quaternion-based algorithms:** Exploring direct quaternion calculation methods like QUEST or Davenport's q-method for improved computational efficiency
3. **Calibration procedures:** Developing systematic calibration methods to account for sensor variations and environmental factors

5.3 System Integration

1. **Earth albedo compensation:** Adding methods to differentiate between direct sunlight and reflected light from Earth, which was identified as a significant issue during testing. One approach would be to test the impact of a secondary light source positioned to simulate Earth albedo and develop filtering algorithms to identify and compensate for this effect.
2. **Power optimization:** Reducing power consumption for space application through duty cycling and more efficient components. Further characterization of the dark current (measured at approximately 0.05 nA with a 3.3V reverse bias, resulting in negligible 5×10^{-7} V across the 10k Ω resistor) can help optimize power requirements.
3. **Flight software integration:** Adapting the algorithms for implementation on flight-qualified hardware. This would include translating the Python-based implementation to a more efficient language like C/C++, as recommended by other team members working on the embedded systems aspects.
4. **Radiation protection:** Adding coverglass to photodiodes to prevent degradation from radiation, as observed in other CubeSat implementations [3]. This would be essential for maintaining sensor performance throughout the mission lifetime.

Additional research is required to determine the expected brightness of the sun in low Earth orbit and to calibrate the sensors accordingly. Furthermore, research into previous literature on dealing with light spillage (equivalent to Earth albedo in space) will be necessary to develop a robust solution to this challenge.

For the next phase of development, it would be beneficial to implement a more advanced photodiode configuration, potentially using the RAX-2 CubeSat approach with three photodiodes on each face to achieve 99.5% coverage of the

attitude sphere [3]. This would improve the system's ability to determine attitude accurately regardless of orientation.

The successful demonstration of this sun sensor system represents an important step toward developing a complete, low-cost ADCS for CubeSat applications. The insights gained from this project will inform future iterations, with the ultimate goal of producing a flight-ready system capable of reliable operation in the space environment.

ACKNOWLEDGEMENTS

I would like to thank the members of the Cooper Union Satellite Launch Initiative (CUSLI) ADCS team for their collaboration and support throughout this project. Special thanks to Adin Sacho-Tanzer for sharing insights from his photodiode characterization testing and for his work on the magnetometer implementation. I also acknowledge the contributions of Marcus Rodriguez, Somin Lee, and Zihan Zhao for their assistance with the mechanical aspects of the project, particularly the enclosure and gimbal design. Additionally, I thank Isabella and Jay for providing access to the Ossila Solar Simulator and their expertise. Finally, I extend my gratitude to Professor Rosen for his guidance and feedback throughout the semester.

REFERENCES

- [1] J. R. Wertz and W. J. Larson, Space Mission Analysis and Design, 3rd ed. Microcosm Press, 1999.
- [2] F. L. Markley and J. L. Crassidis, Fundamentals of Spacecraft Attitude Determination and Control. Springer, 2014.
- [3] Springmann, J. C., and Cutler, J. W., "Flight Results of a Low-Cost Attitude Determination System," *Acta Astronautica*, Vol. 99, 2014, pp. 201-214.
- [4] R. B. Sharma and A. Tewari, "Optimal Nonlinear Tracking of Spacecraft Attitude Maneuvers," *IEEE Transactions on Control Systems Technology*, vol. 12, no. 5, pp. 677-682, 2004.
- [5] M. D. Shuster and S. D. Oh, "Three-Axis Attitude Determination from Vector Observations," *Journal of Guidance and Control*, vol. 4, no. 1, pp. 70-77, 1981.
- [6] Sinclair, D., Grant, C. C., and Zee, R. E., "Enabling Reaction Wheel Technology for High Performance Nanosatellite Attitude Control," 21st Annual AIAA/USU Conference on Small Satellites, 2007.
- [7] Sacho-Tanzer, A., "CubeSat Attitude Determination System Design," Unpublished manuscript, Cooper Union, New York, 2025.